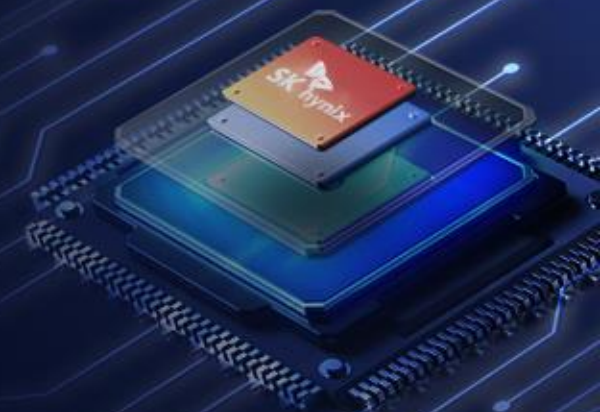




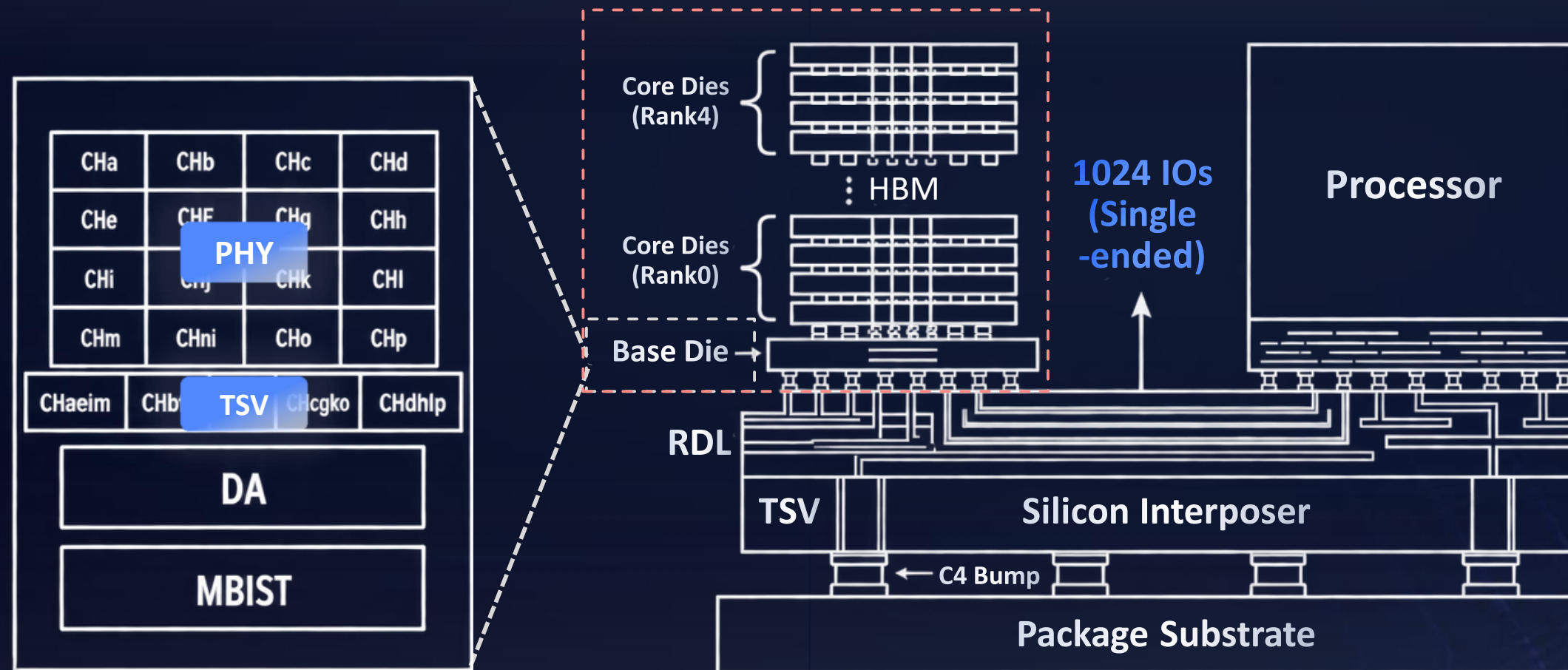
Advanced Packaging for High Bandwidth Memory (HBM)

Jaesik Lee
SK hynix America



HIGH BANDWIDTH MEMORY AND ADVANCED PACKAGING

- HBM is a 3D-stacked structure, which has a base die and a stack of core dies, up to 16 slices
GPU and HBM, mounted on Si interposer, are communicating through 1024 IOs with 16 channels



- HBM is a 3D-stacked structure by using TSVs
 - A base die with multi-core dies up to 16 slices
 - 4 slices per rank, 4 ranks/16Hi
 - 4 channels per slice, 16 banks
- HBM and GPU are mounted on a Si interposer by 2.5D package
- 1024 IOs with 16 channels via Si interposer

WHY HBM?

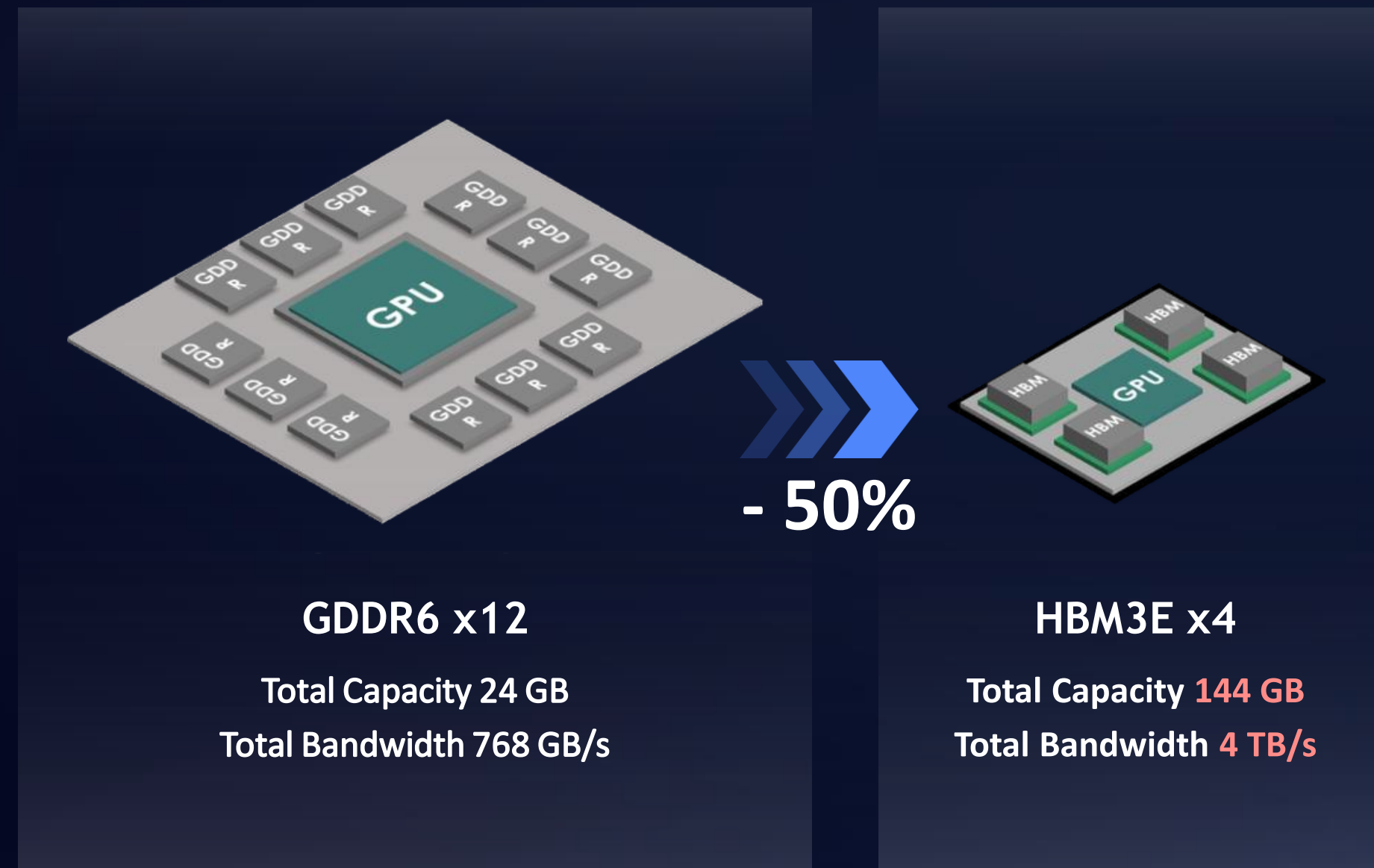
REQUIREMENTS FOR MEMORY

- ✓ **Higher Bandwidth**
to support multi-processing of accelerator/GPU
- ✓ **Higher Capacity**
for huge data size and parameters for generative AI
- ✓ **Higher Power Efficiency**
for power & cooling management of data center

HBM BENEFITS

- The more HBM adoption is more save the space, power & operation cost

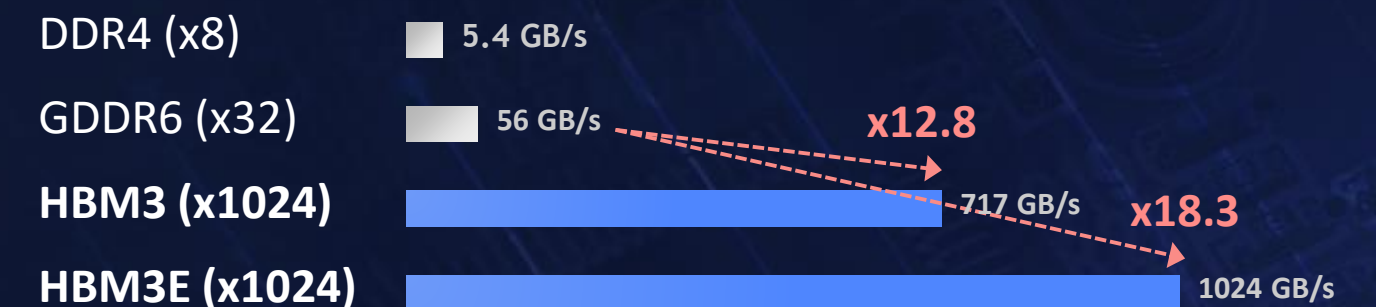
1.Highest Space Efficiency



2.Highest Memory Capacity



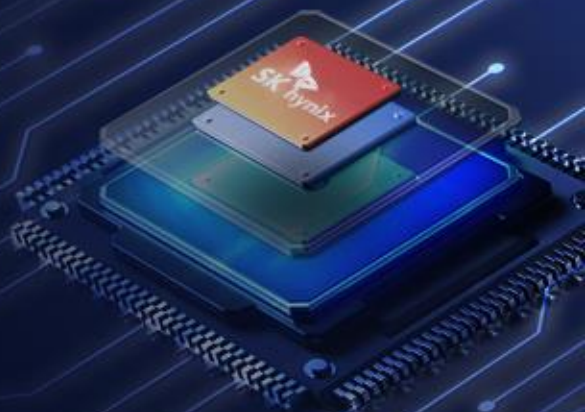
3.Highest Memory Bandwidth



4.Highest Power Efficiency



HBM ROADMAP

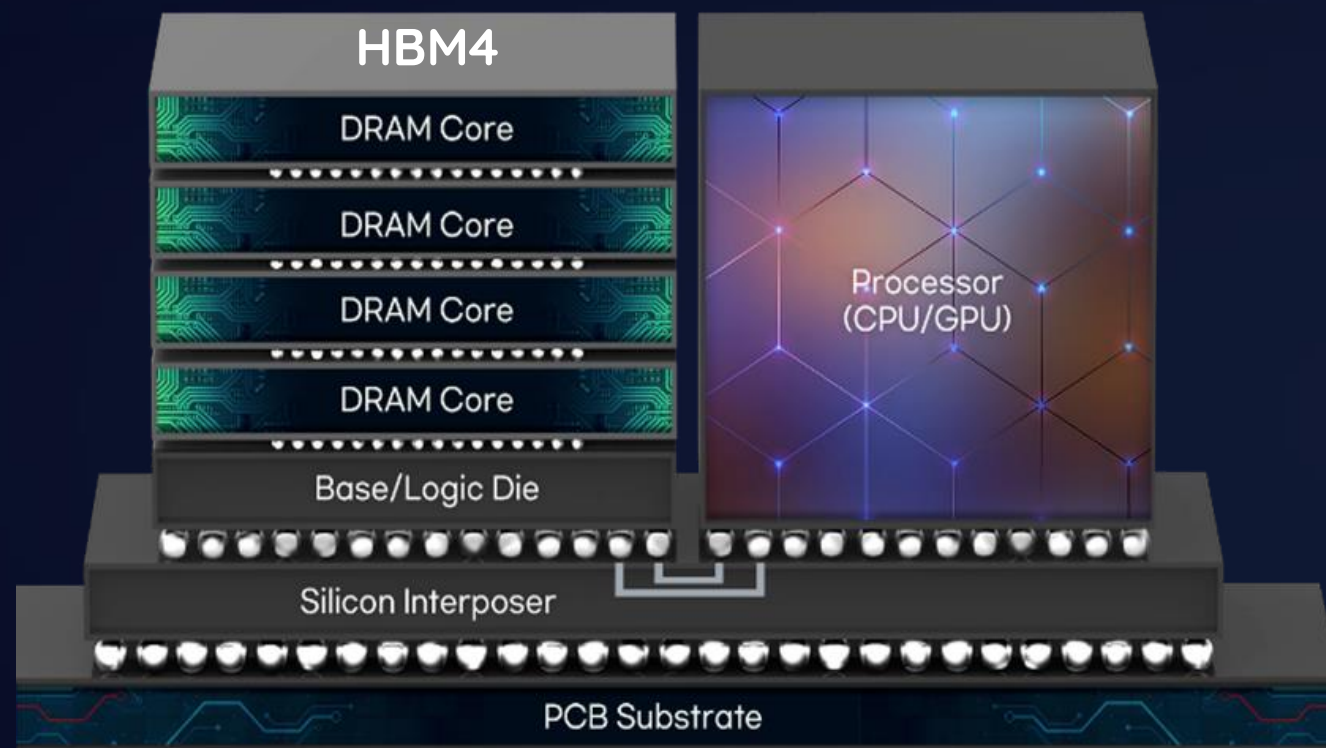


HBM Capacity & Bandwidth Increases vs HBM Generation

		HBM2E	HBM3	HBM3E	HBM4
Capacity	Density (Gb)	16	16	24	24
	Capacity	16GB	24GB	36GB	36GB
Perf.	Total I/O	1024	1024	1024	2048
	IO Speed (Gbps)	3.6	5.6	8	8
	Max BW (GB/s)	460	717	1024	2048
Package	PKG size (mm ²)	10x11	11x11	11x11	12.4x11

HBM4 OVERVIEW

- HBM4 package height and size are bigger and implements more TSV and uBumps than HBM3E

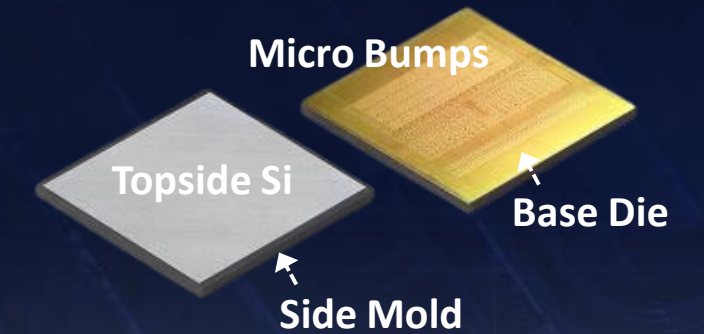


- >2TB/S BW, 40+% lower power efficiency
- 14+% Improving thermal resistance than HBM3E
- Up to 48GB Capacity (12Hi in Production, 16Hi under Qual)
- 775um Z-Height, 12.8x11mm²
- 16148 Base Micro-Bumps, >20K TSVs

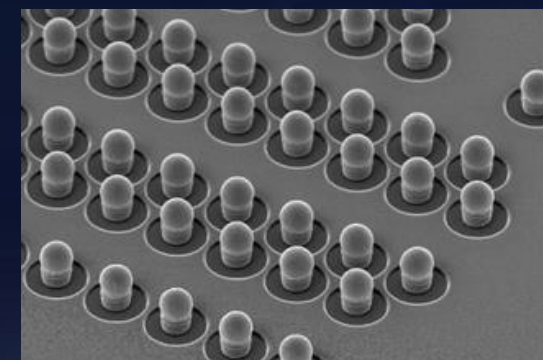
HBM Cubes



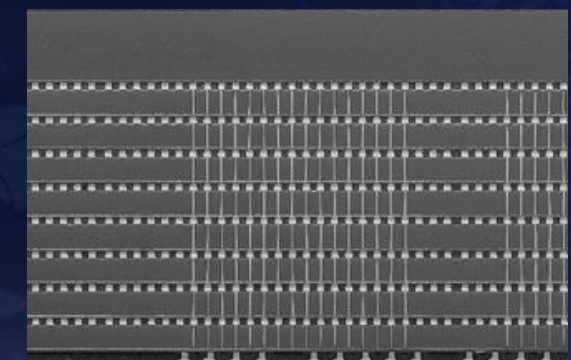
Top/Bottom View



Base Die Bump



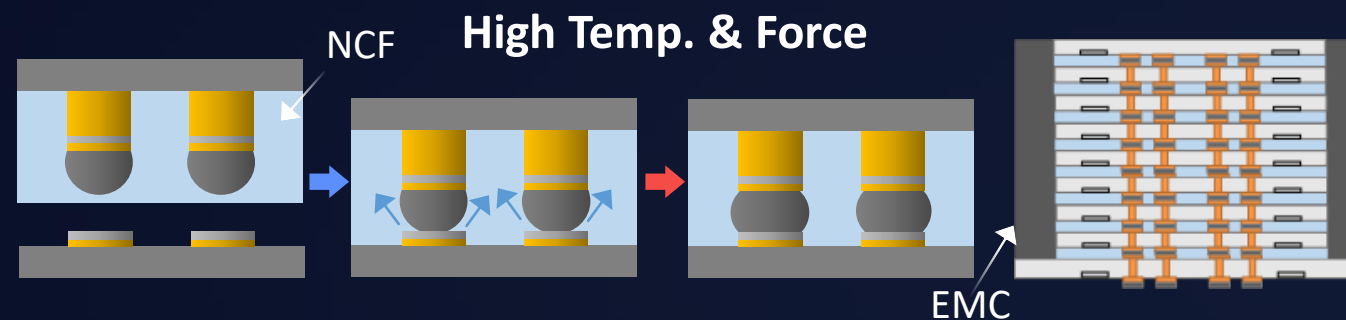
TSV/Bump Joints



HBM PACKAGE TECHNOLOGIES

TC+NCF

(Thermo-compression + Non-Conductive Film)



PROCESS

Thermo-compression bonding with NCF

PROS

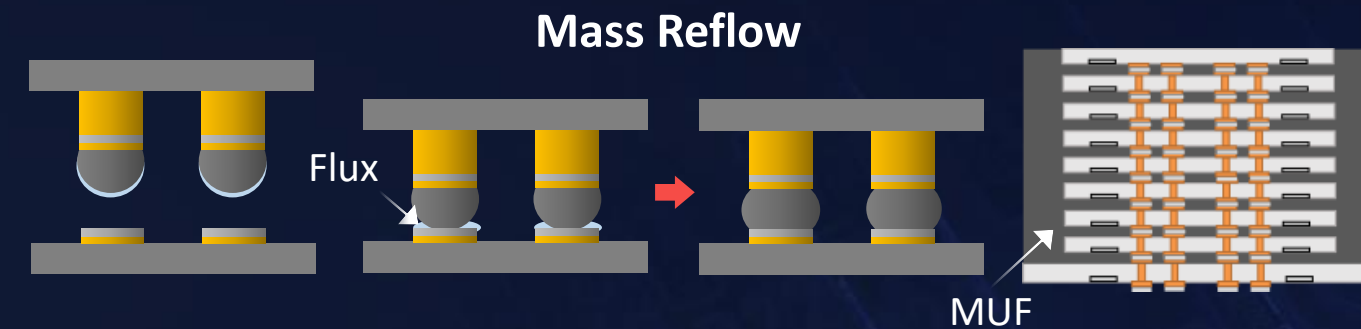
- Less sensitive to Thin Die Warpage

CONS

- Low productivity (each die bonding)
- High thermal resistivity

MR+MUF

(Mass Reflow + Molded Underfill)



PROCESS

Mass Reflow + Molded underfill (MUF)

PROS

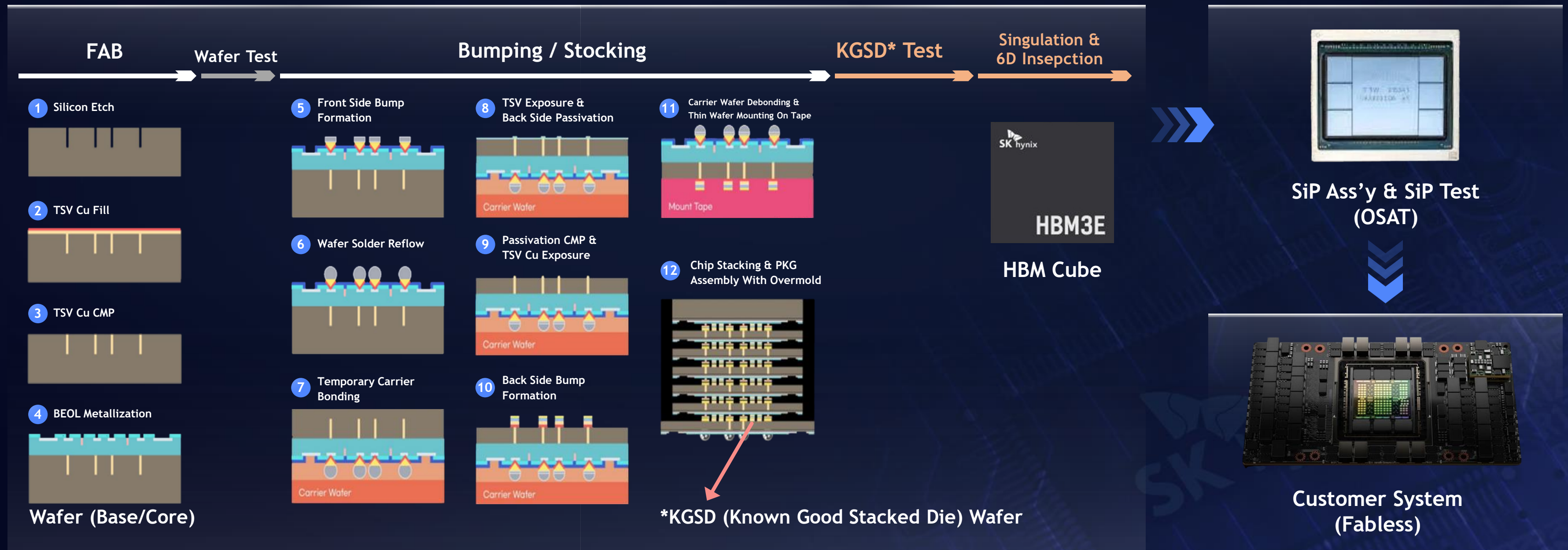
- High Productivity
- Low thermal resistivity

CONS

- Sensitivity to Chip Warpage
- Narrow Gap Fill

HBM PROCESS FLOW, SK HYNIX

Wafer (Base/Core) >>> WT >>> WLP & KGSD (Known Good Stacked Die) Wafer >>> KGSD Wafer Test
>>> Singulation & 6D Inspection >>> HBM Cube Release



SK HYNIX'S HBM PACKAGE, KEY TECHNOLOGIES

- TSV formation, micro-bumping, wafer thinning, and chip stacking are key features for HBM packaging

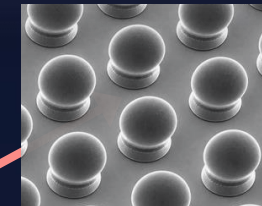
Via (TSV) Formation

- Process : Si Via Etch, Liner/BM Dep, High AR Cu Filling
- Concern : KOZ, IMD Integrity, Cu Contamination



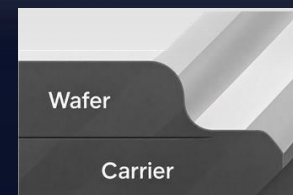
u-Bump Formation

- Process : Electroplating
- Concern : Process Uniformity, Yield, Reliability (Mechanical/Thermal)

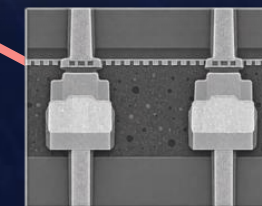


Wafer Thinning

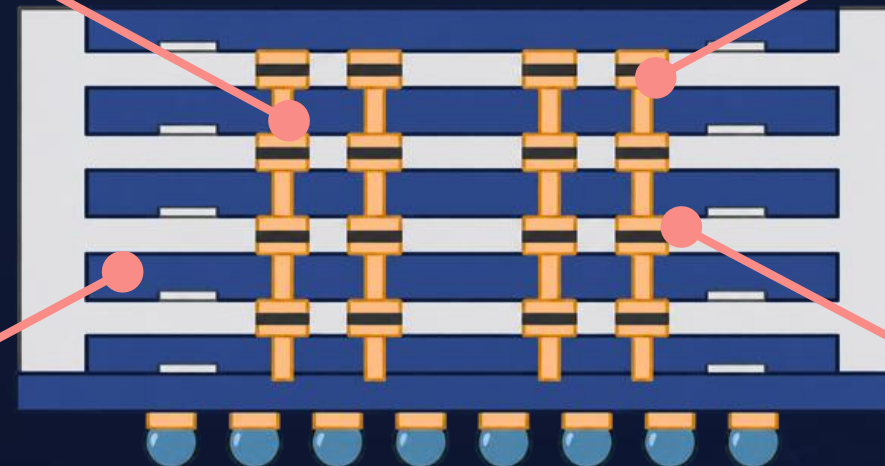
- Process : Temporary wafer bonding/debonding, Thinning/Via Reveal
- Concern : TTV Control, Adhesive Residue, Throughput, Backside Passivation



Chip Stack/Underfill



- Process : Post Gap-fill (MUF)
- Concern : Thin Die Handling, Warpage Control, Reliability (Void, Adhesive), Joint Reliability

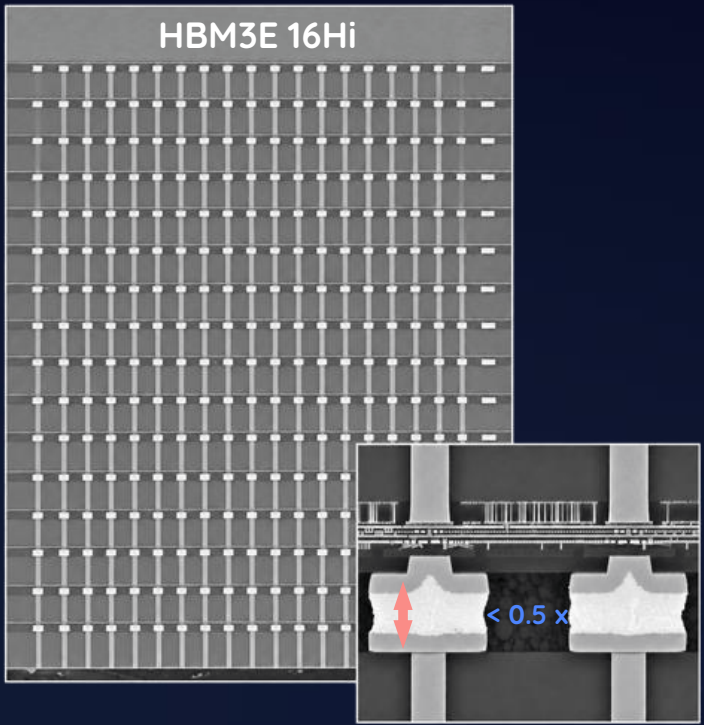


ADVANCED MR-MUF FOR 16HI

- HBM3E 16Hi(@48GB/Cube) is successfully developed by advanced MR-MUF technology

Key Technology

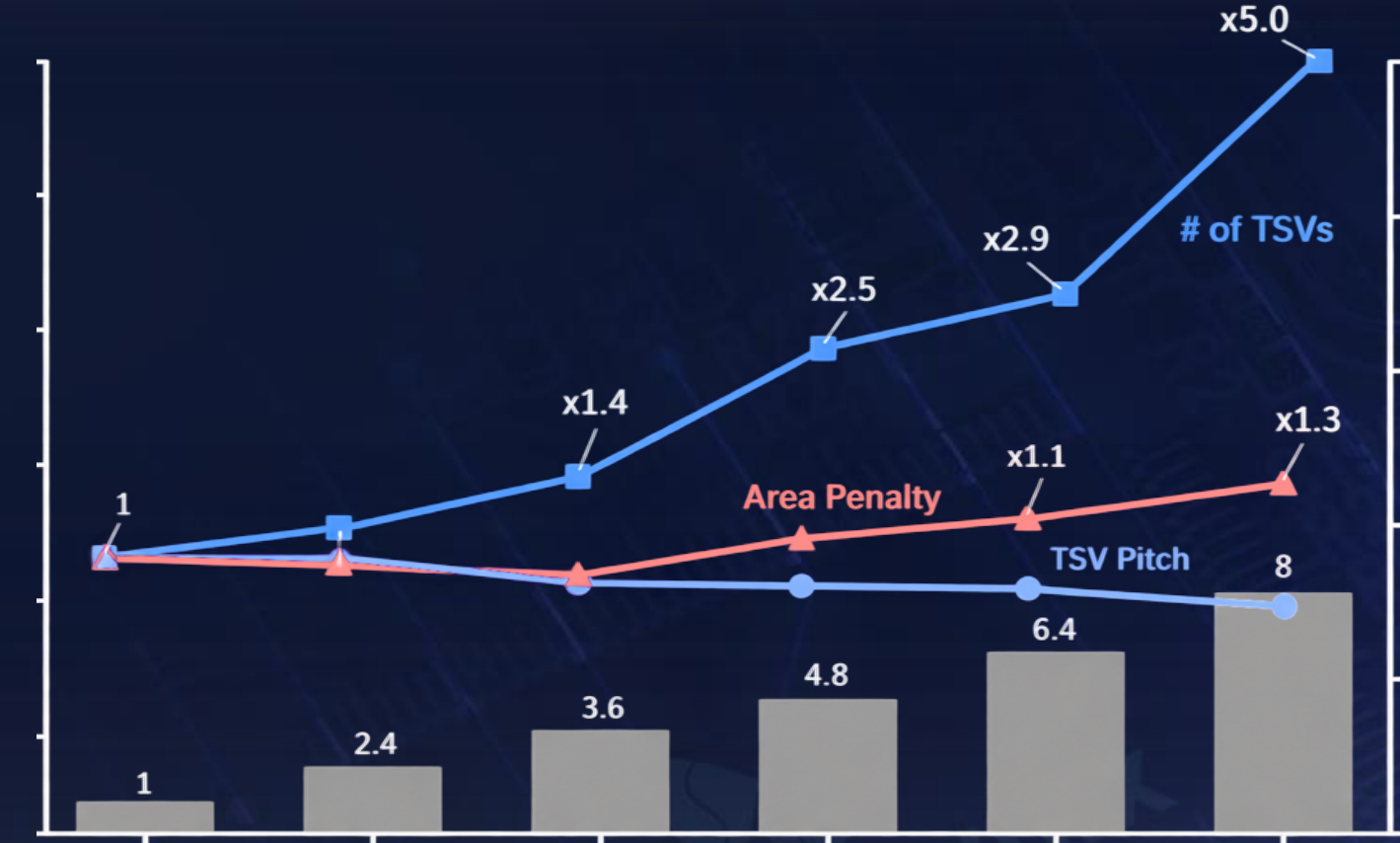
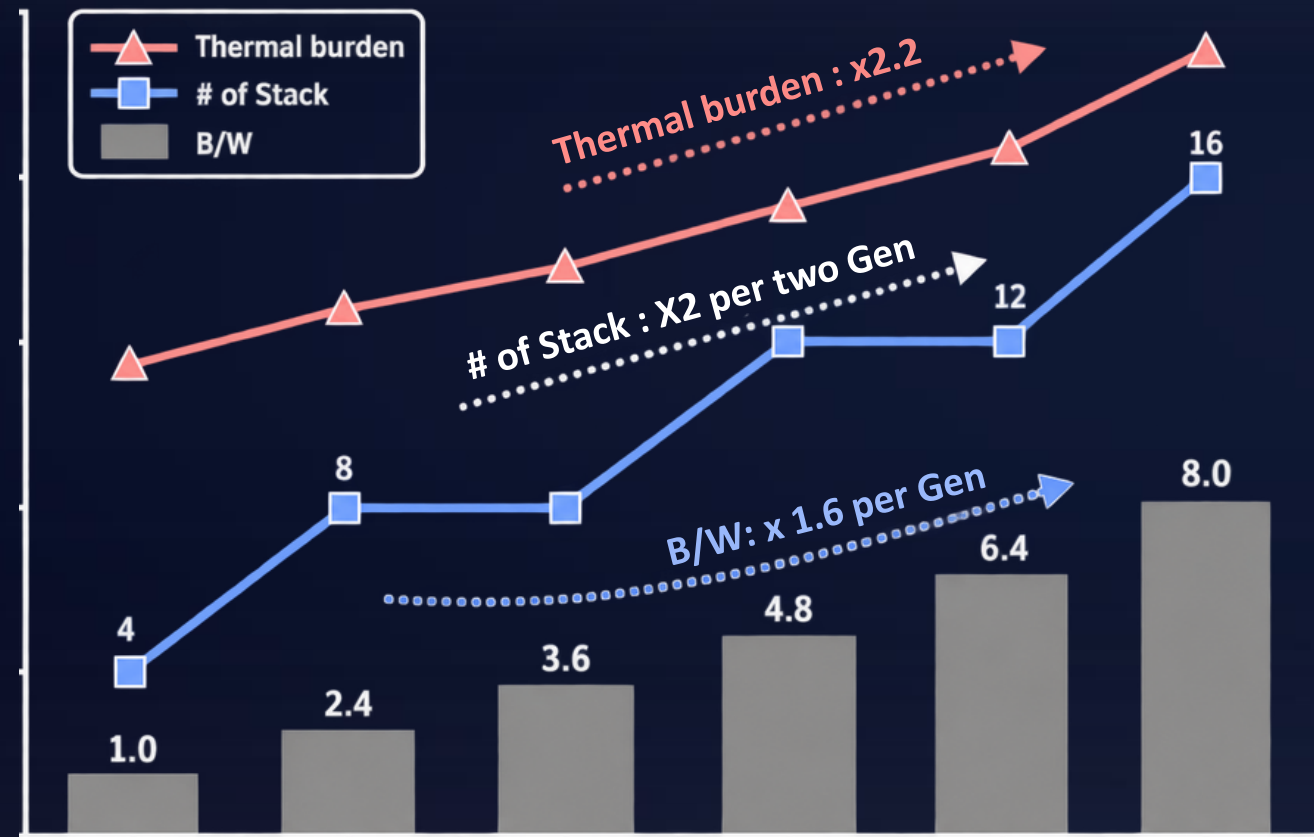
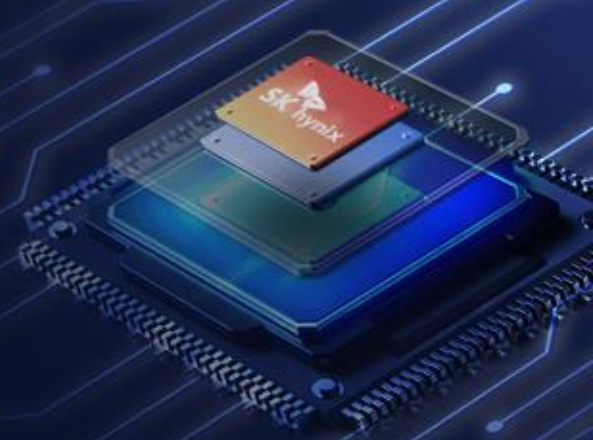
Warpage control, Fine Pitch Int'n & Narrow Gap-fill



Items	12Hi → 16Hi	Challenge
① Total PKG Height (μm)	720 → 775	
② Chip Thickness (μm)	1.0x → 0.9x	Die Warpage
③ Gap-Height (μm)	1.0x → 0.5x	Gap-Fill Quality
④ Bump Pitch (μm)	1.0x → 0.9x	

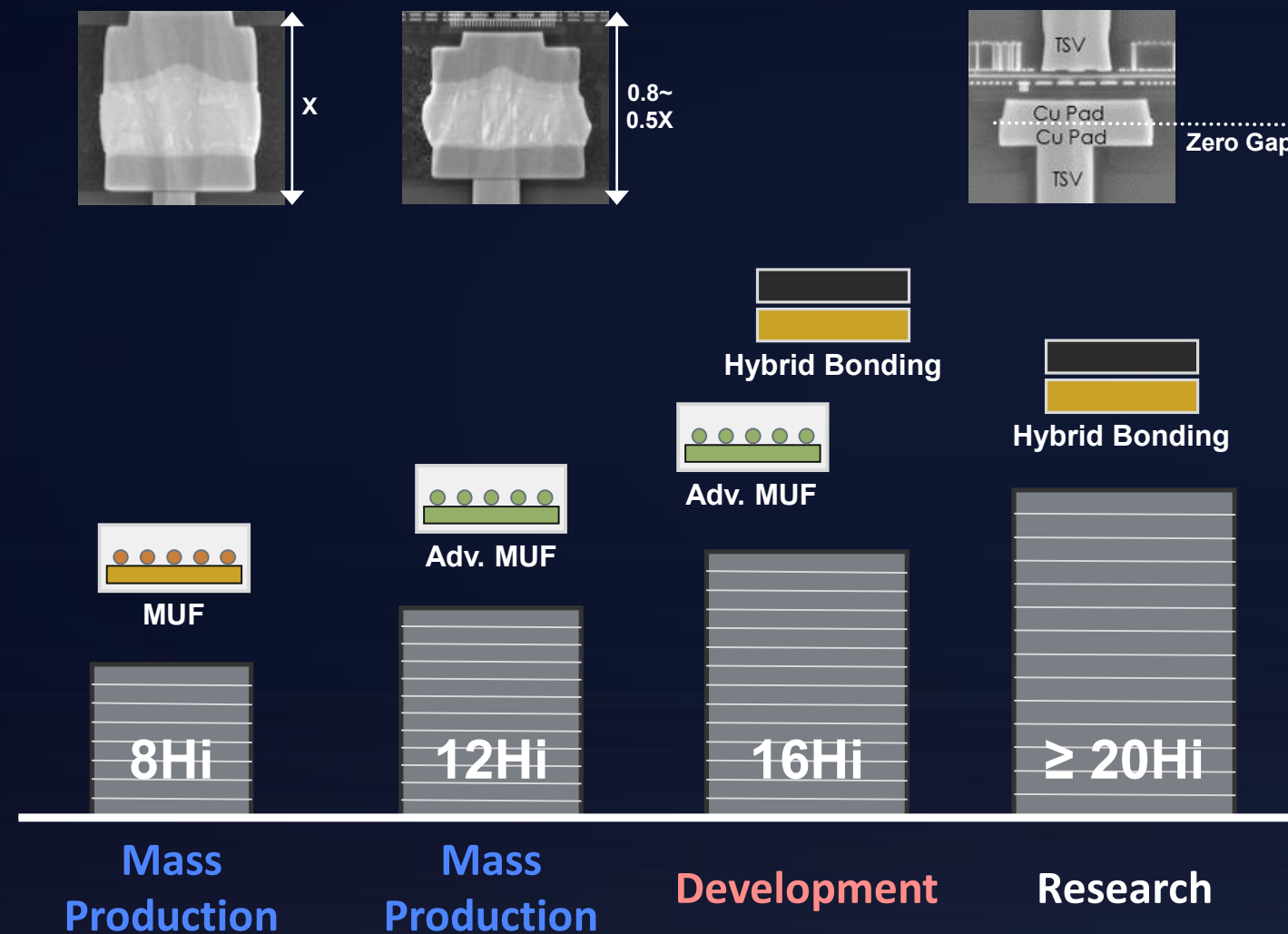
HBM CHALLENGES

- HBM Power & Bandwidth ↑ Thermal Issue & TSV Area ↑

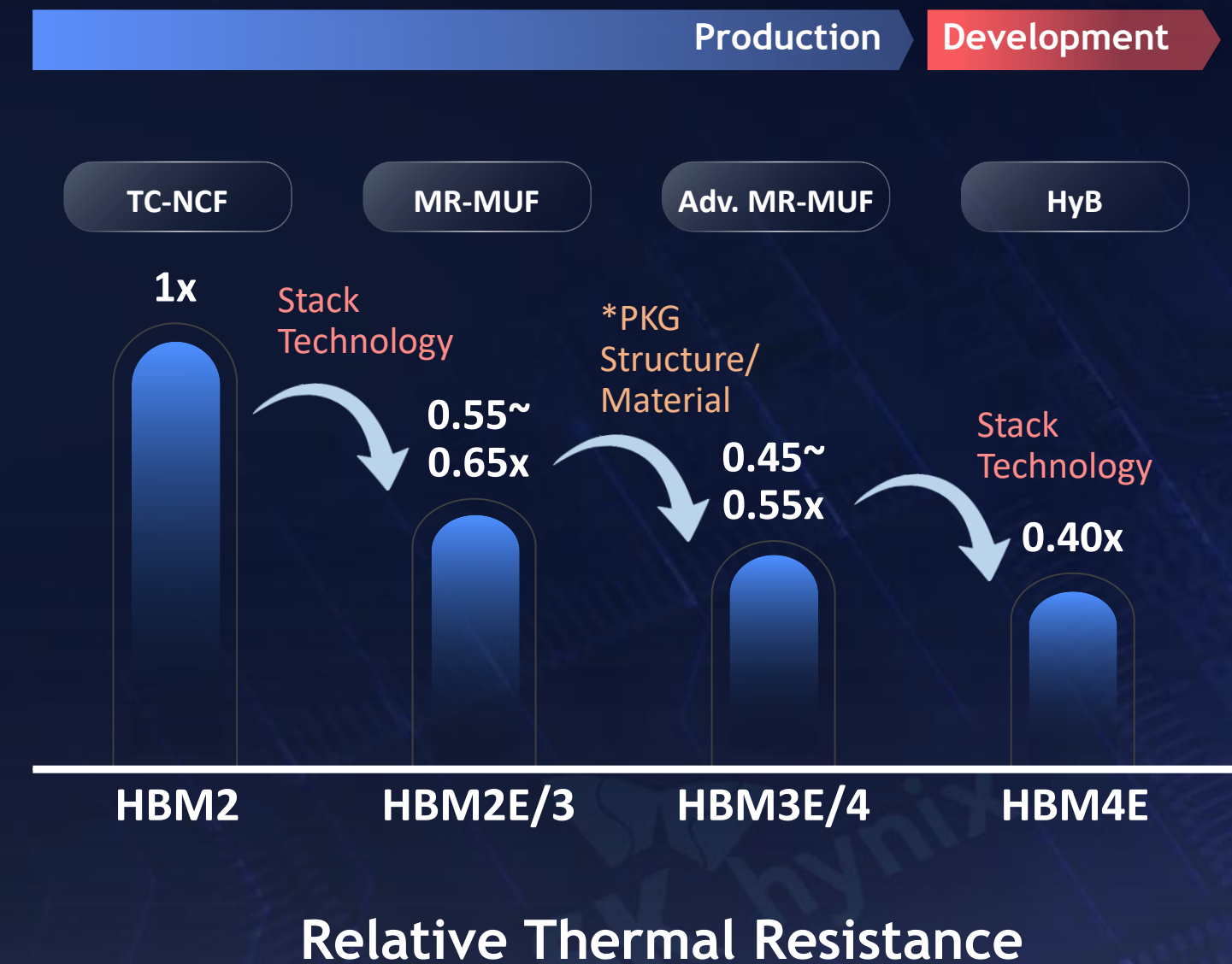


HBM PACKAGE : HBM PACKAGING ROADMAP & CHALLENGES

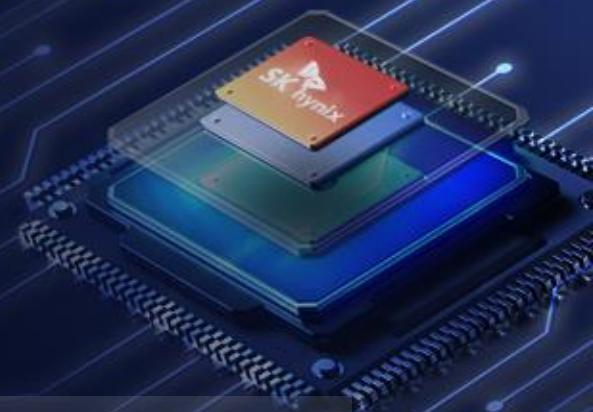
Memory Capacity



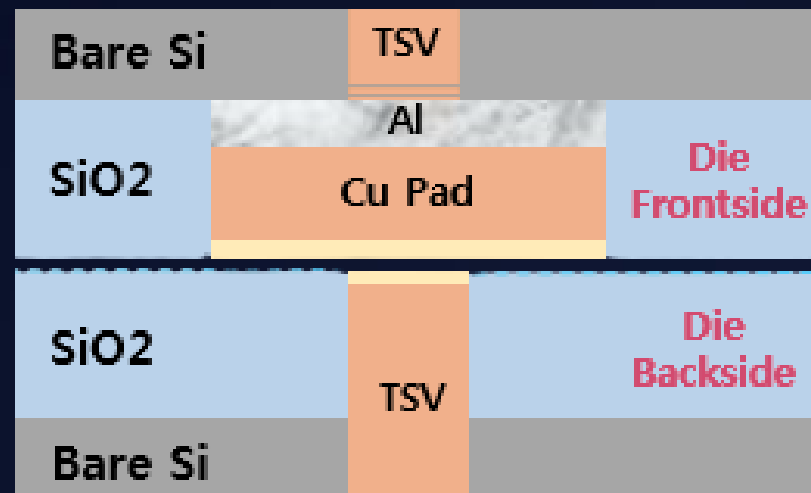
Thermal Management



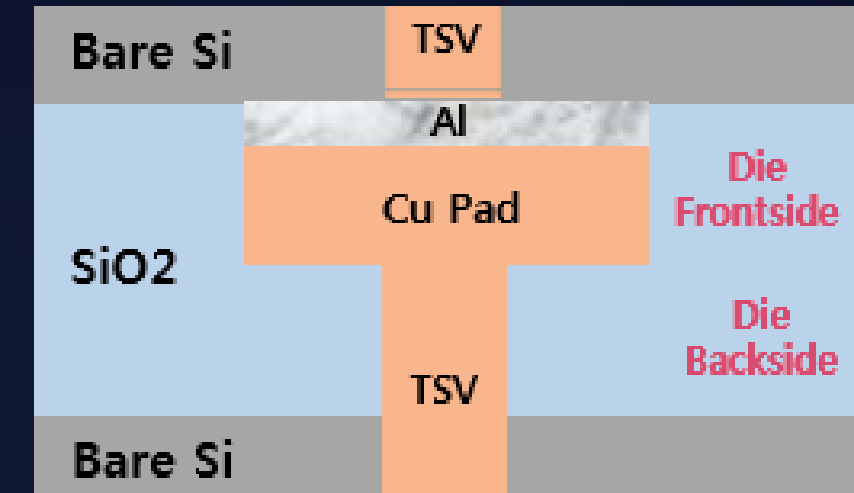
NEXT GEN. HBM PACKAGING, HYBRID BONDING



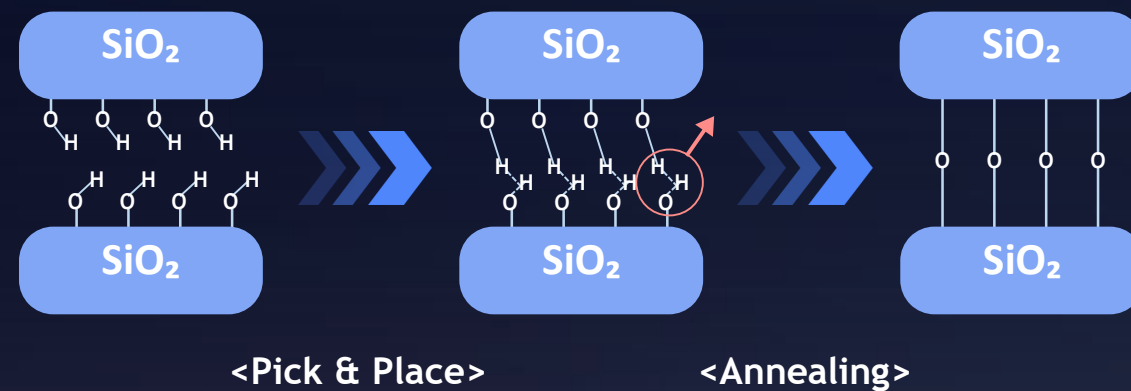
Pick & Place @ Room Temp.



Annealing > 200°C



SiO₂-SiO₂ Bonding



Cu-Cu Bonding



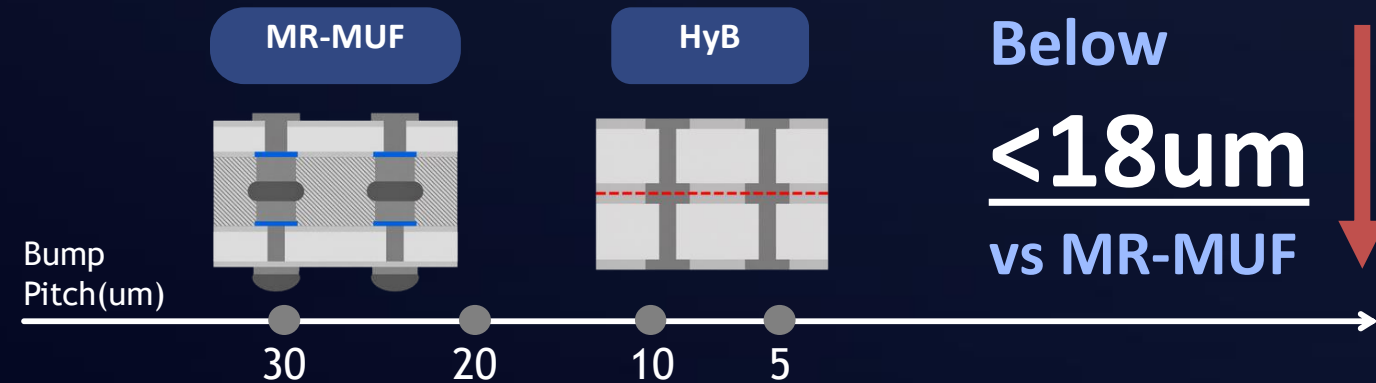
NEXT GEN. HBM PACKAGING, HYBRID BONDING

- Hybrid bonding is a promising more memory capacity ($\geq 20\text{Hi}$), more performance (narrow pitch) and more thermal efficiency (high conductivity)

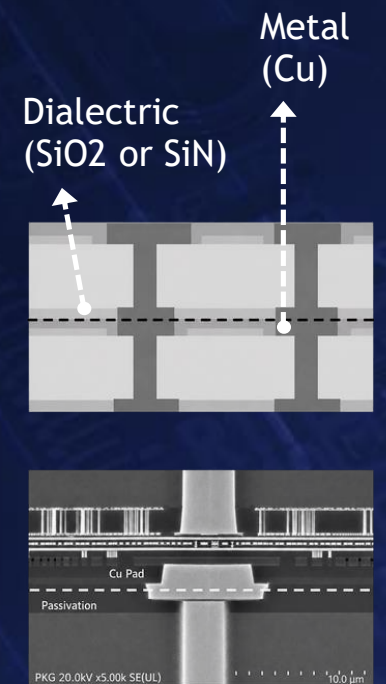
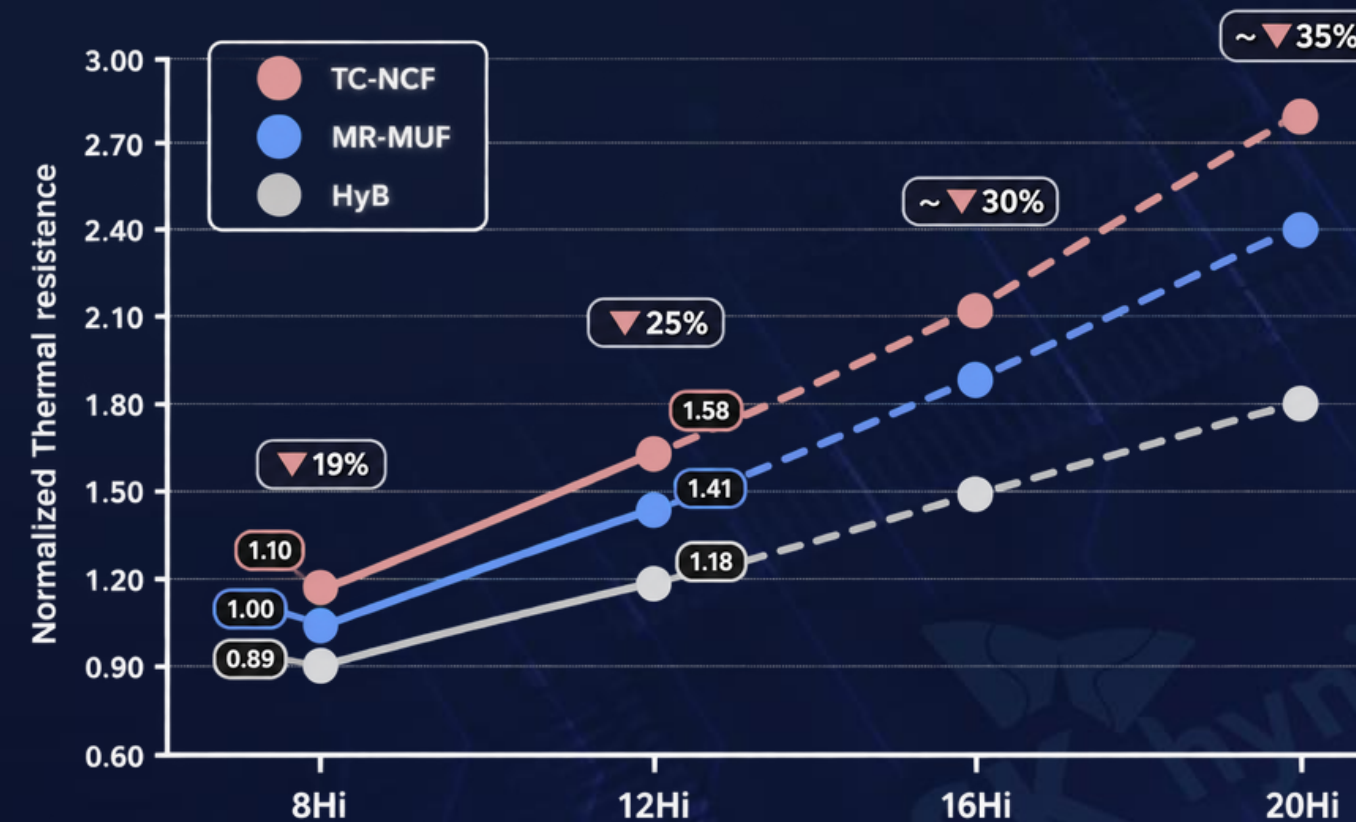
1 Under limited Z-height, HyB provides thicker core die



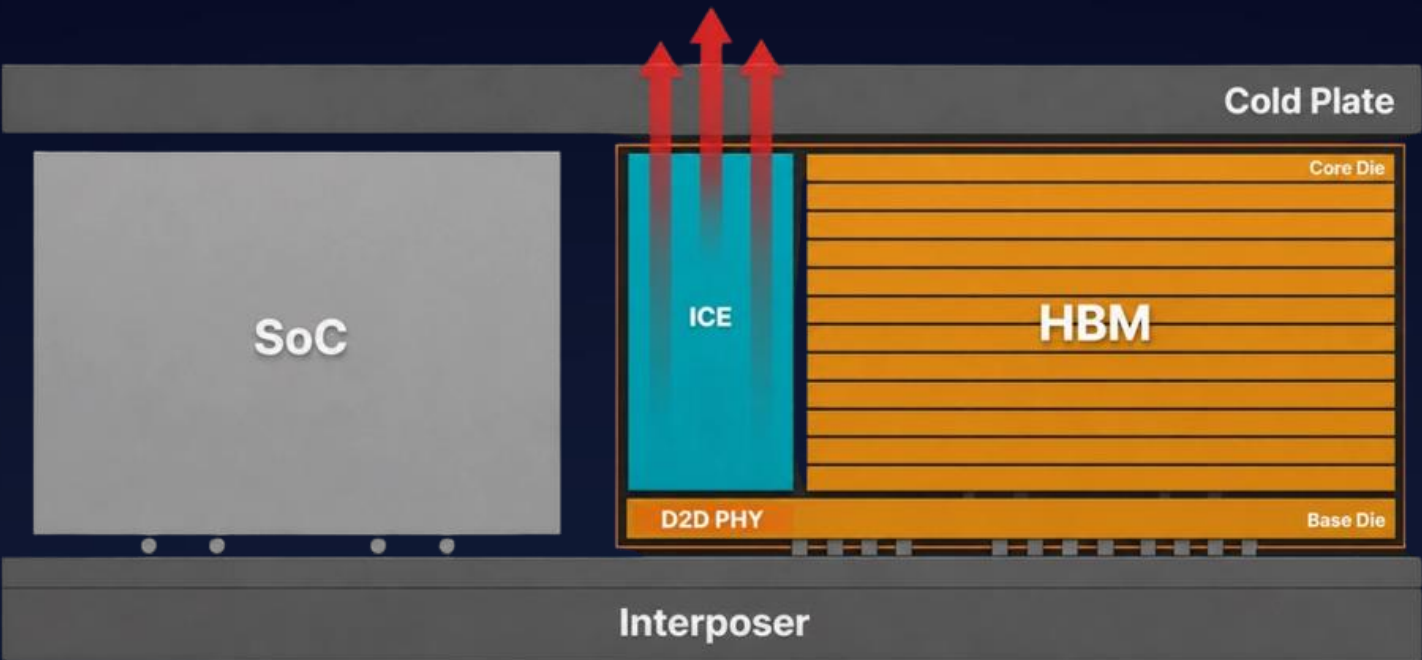
2 HyB enables narrow TSV pitch



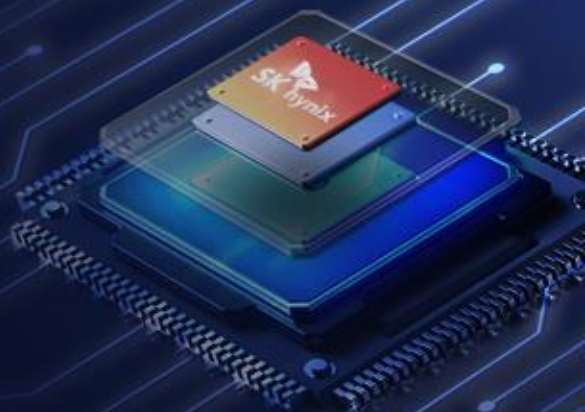
3 More thermal efficiency with more stack layers



I-HBM Concept: Localized Hotspot Mitigation



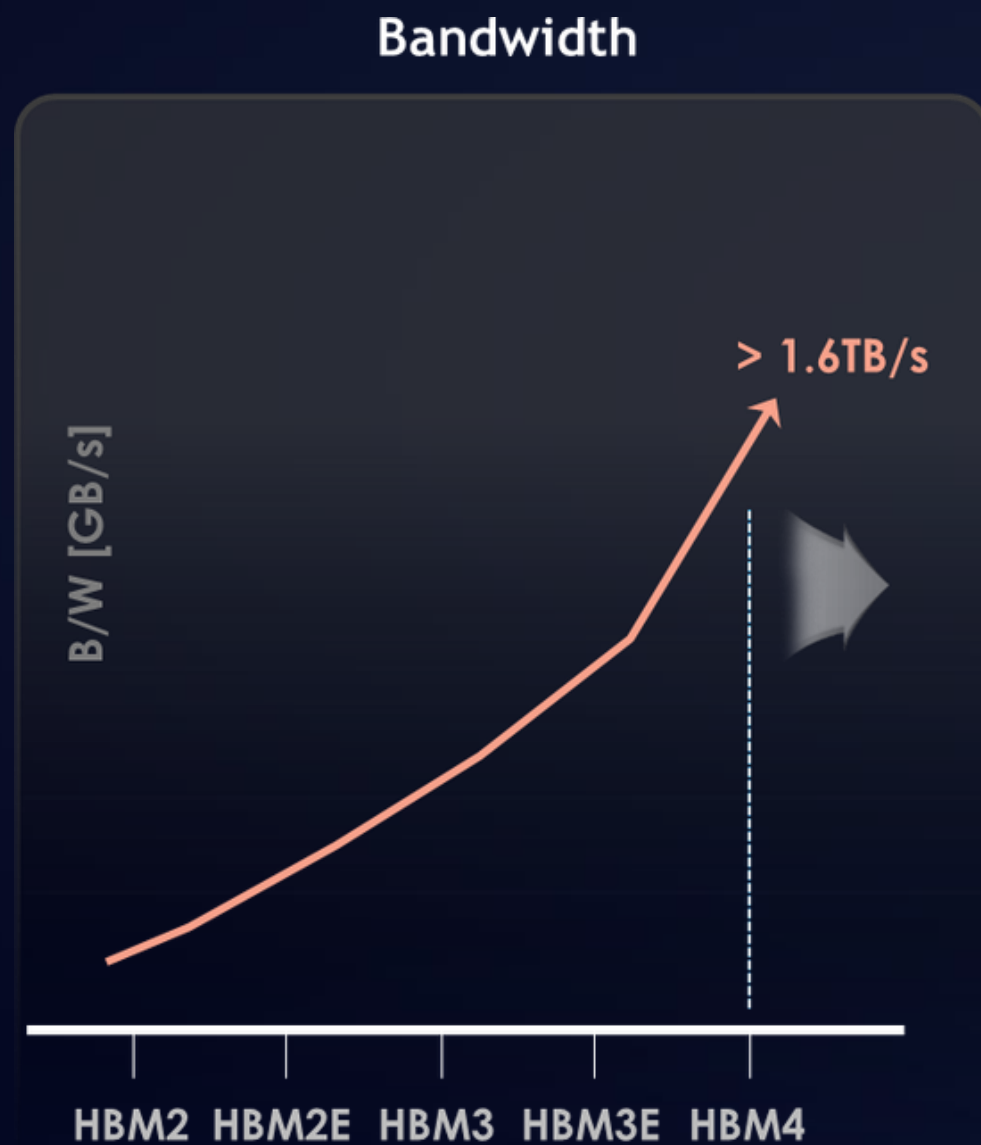
[SK hynix i-HBM]



Company	Technology	Core Idea	Key Data
SK Hynix	iHBM	Embed a high-thermal-conductivity, electrically insulating cooling component inside the HBM's hot D2D PHY area, creating a dedicated heat path	>30% reduction in thermal resistance
Samsung Electronics	HPB cooling + hybrid copper bonding	Change chip stacking structure by moving DRAM to the side of the processor and placing a copper heat spreader directly above the processor core; use copper-to-copper bonding to eliminate thermal resistance	~30% temperature reduction; 16% improvement in thermal impedance
Micron Technology	Circuit design improvement + enhanced base die	Improve cooling while boosting performance by refining internal circuit design and optimizing base die performance	>20% improvement in energy efficiency

CHALLENGES FOR FUTURE HBM

- Exploring new solutions to overcome challenges in bandwidth and power consumption for HBM4 and beyond



✓ Proposed Solutions

1. Increasing Bandwidth

- Expand Data IOs
- Increase Speed/IO

2. Better Power Efficiency

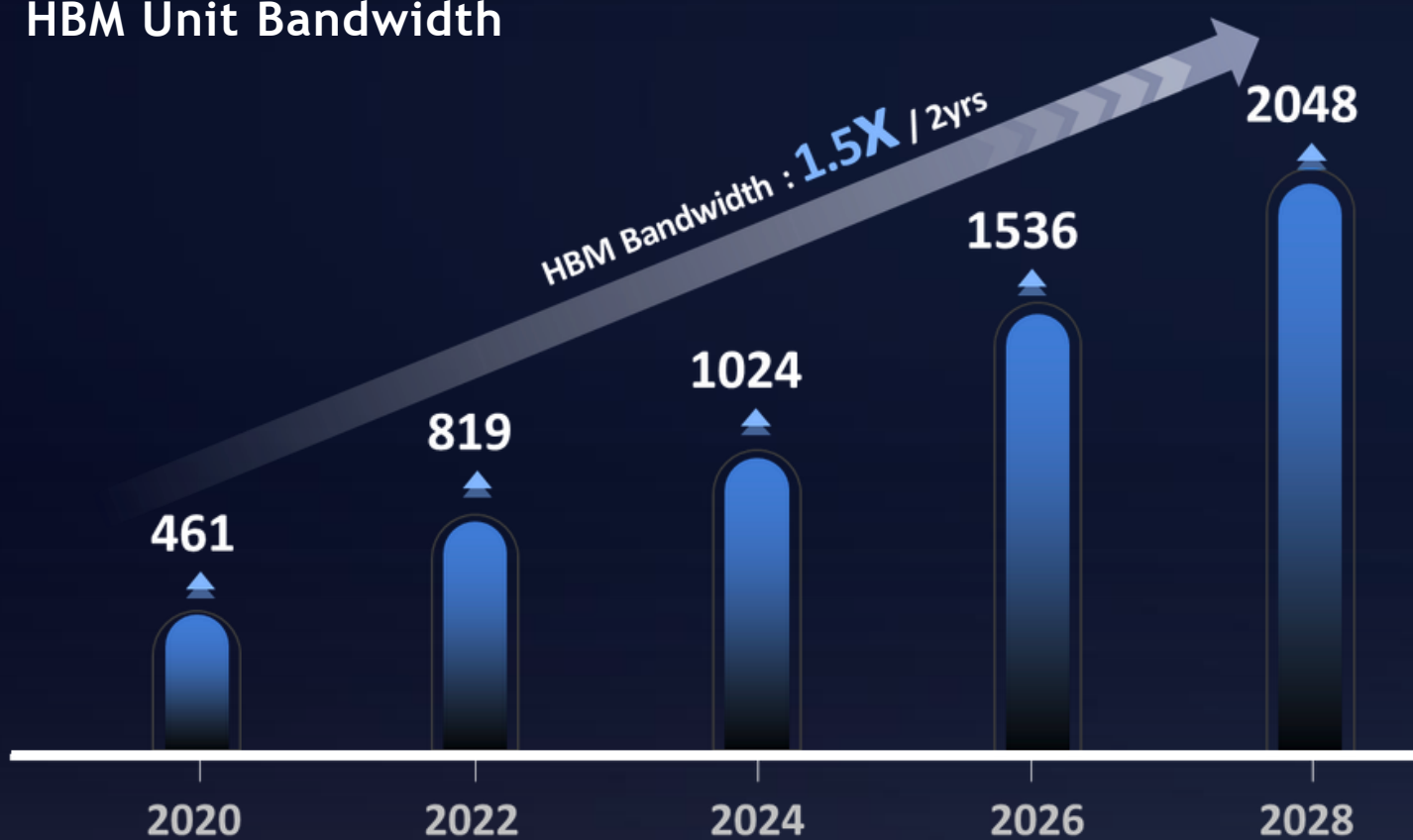
- Logic Basic Die
- Increase # of TSV

CHALLENGES FOR FUTURE HBM, BANDWIDTH CHALLENGE

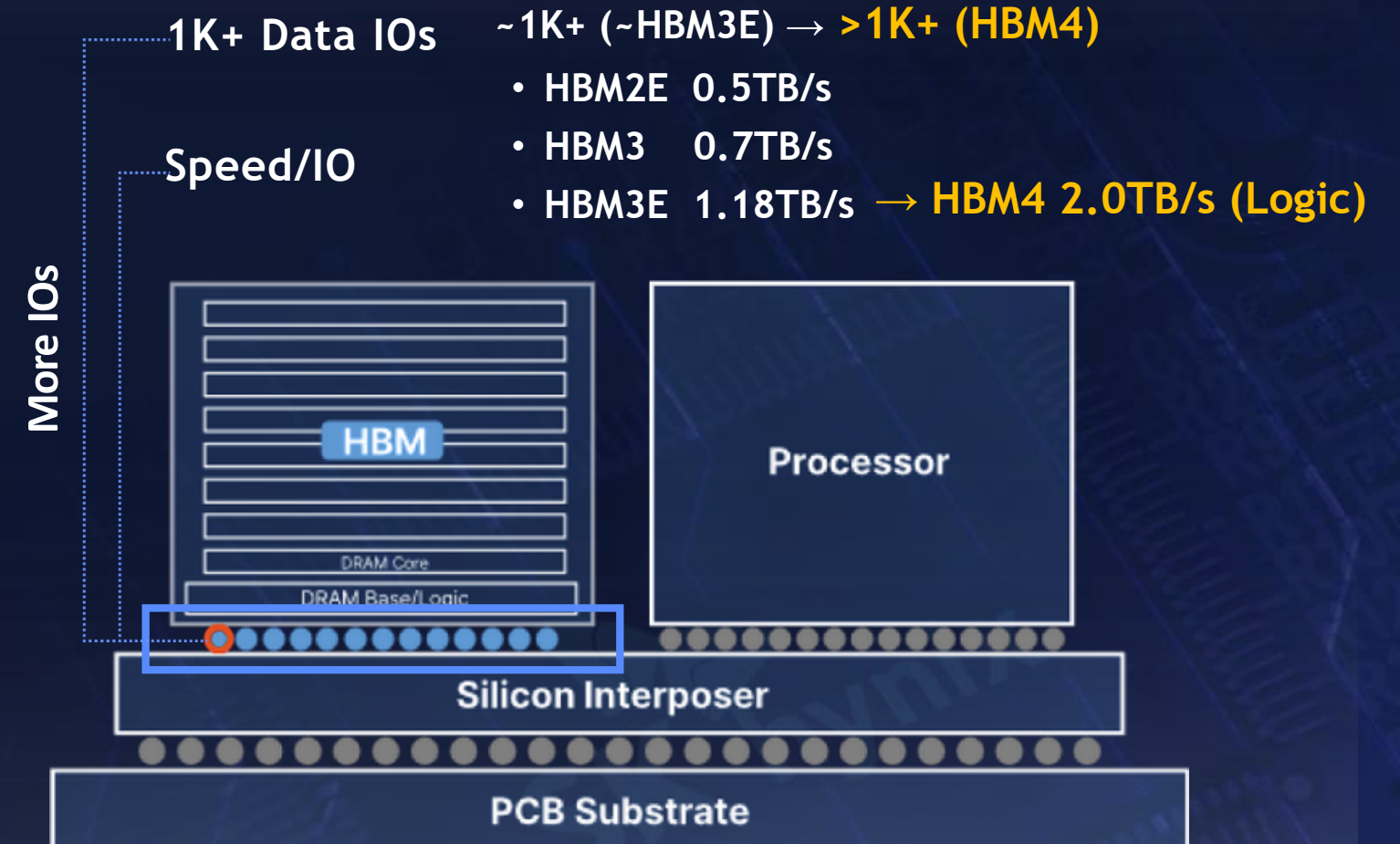
- Enhance bandwidth by more data I/Os (expansion of data-rate/IO) with doubled TSV number and higher IO speed with logic process integration

Bandwidth Demand

HBM Unit Bandwidth



Solution

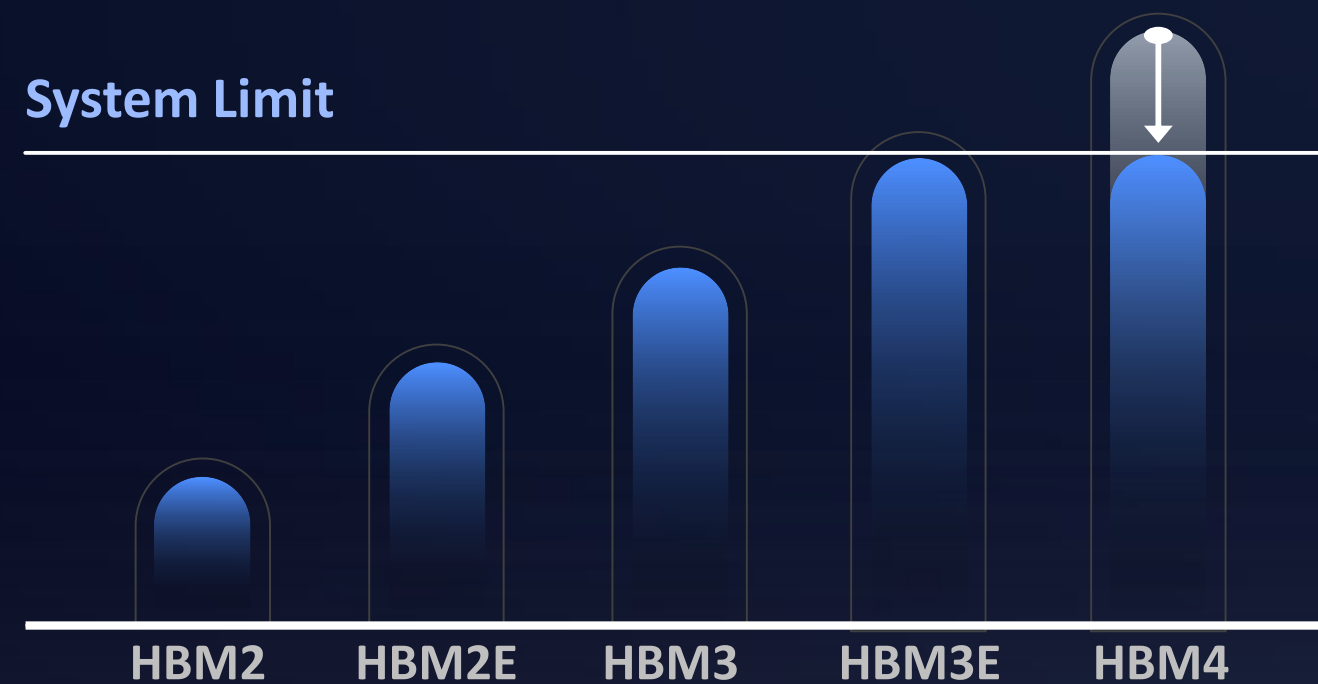


CHALLENGES FOR FUTURE HBM, POWER/PDN CHALLENGE

- Dramatic power efficiency improvement by optimized logic foundry process
- Power TSVs spread “everywhere” greatly improves PDN

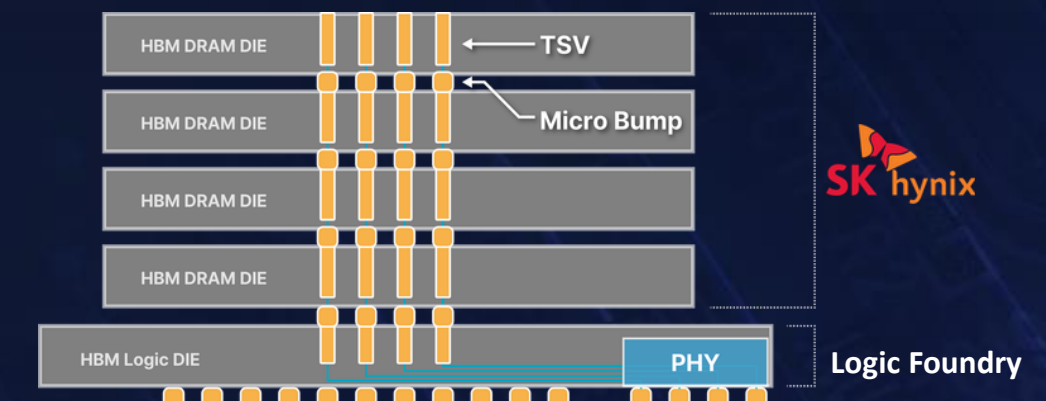
Power Trend

System Limit

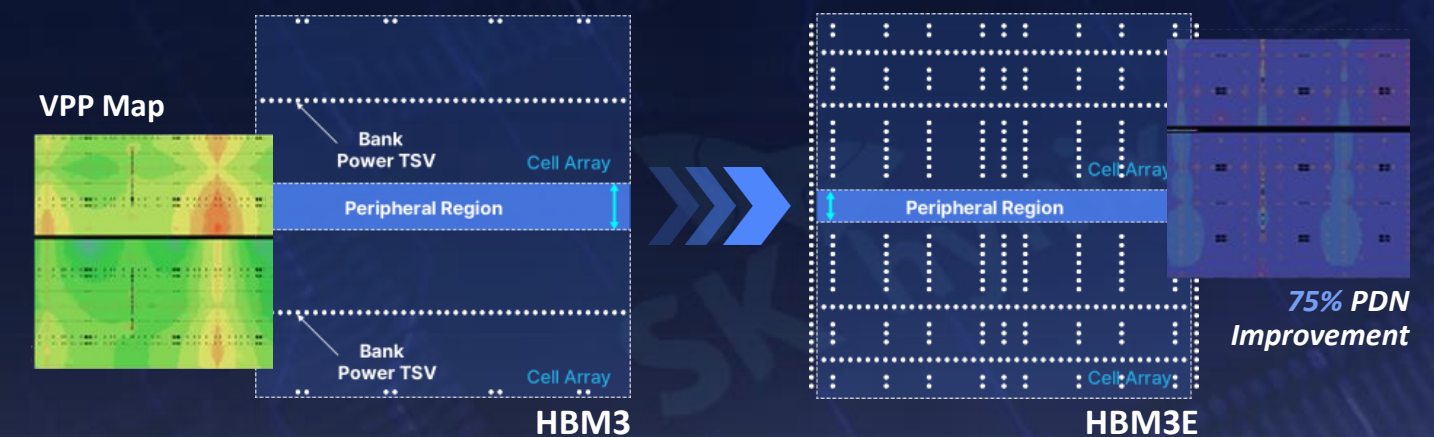


Solution

- Advanced Logic Process for Low Power



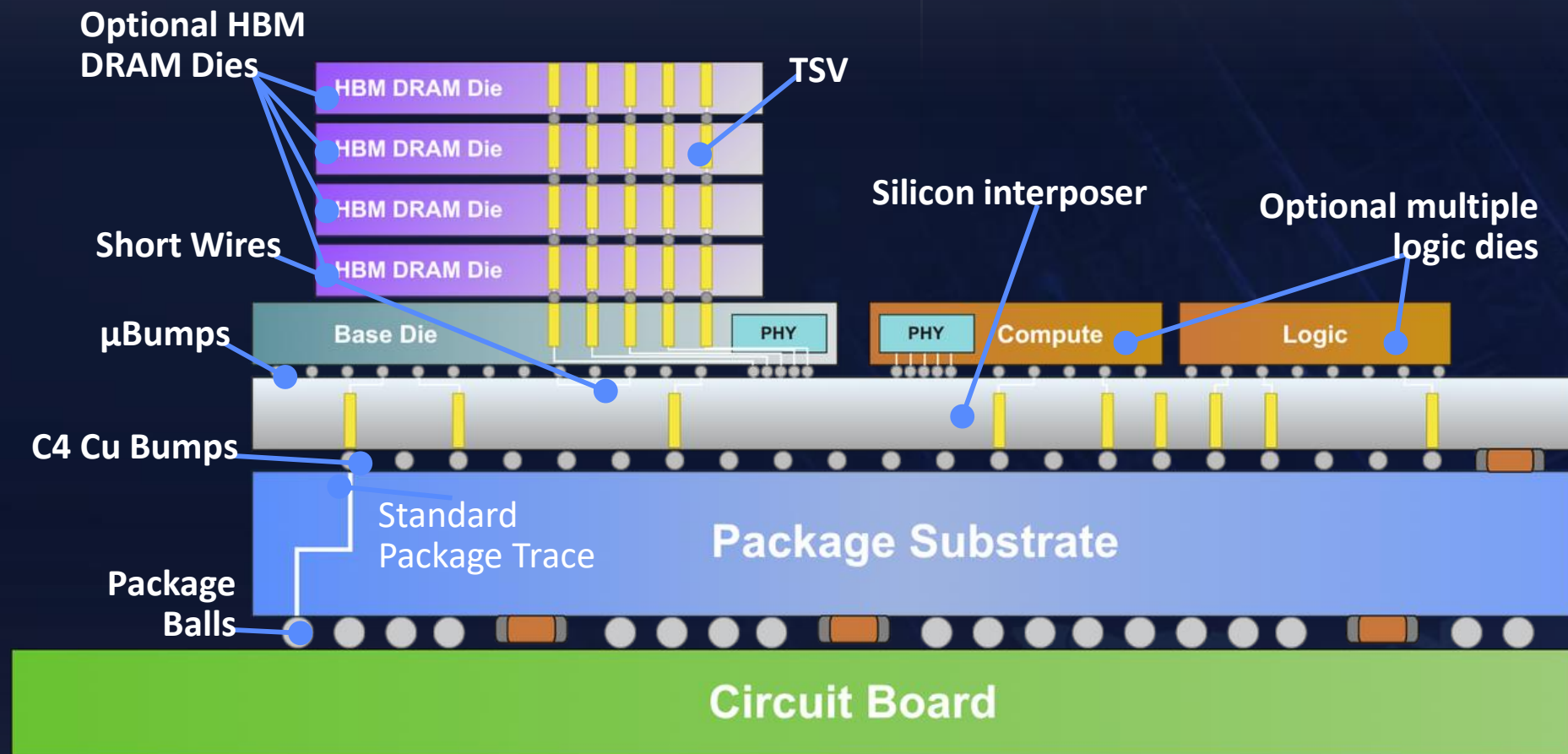
- Power TSVs spread “everywhere” greatly improves PDN



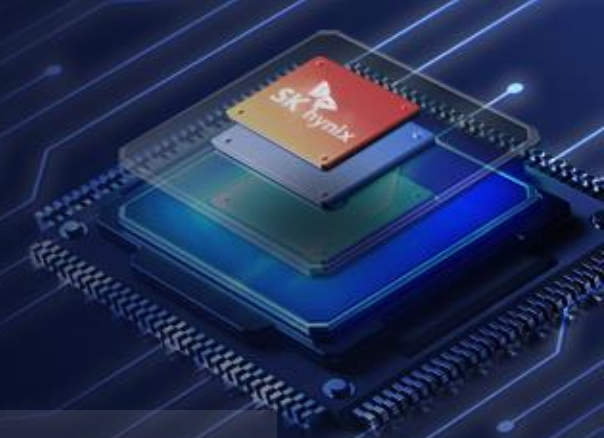
2.5D PACKAGING, **IMPACT ON HBM**

- In the past, Memory was assembled at the last step of system integration
- In Adv PKG for AI, HBM is assembled at the first step leading to significant reliability challenges

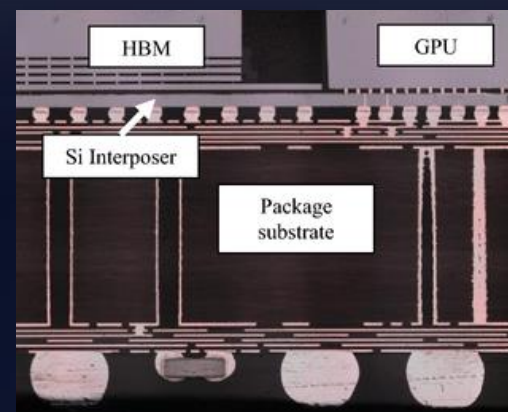
Intel® Whitely Platform Server Board



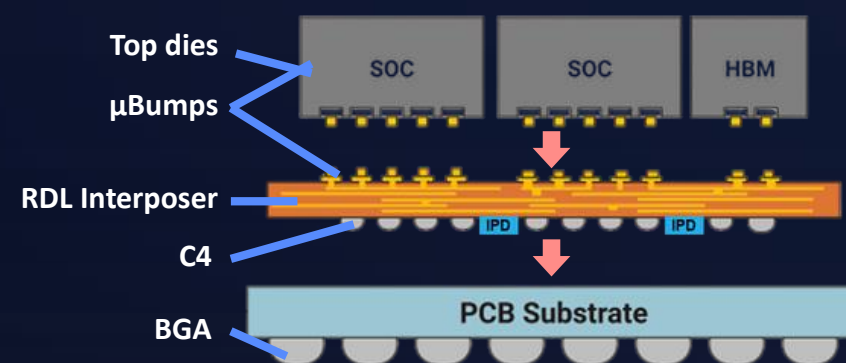
2.5D PACKAGING, **IMPACT ON HBM**



CoWos-S



CoWos-R



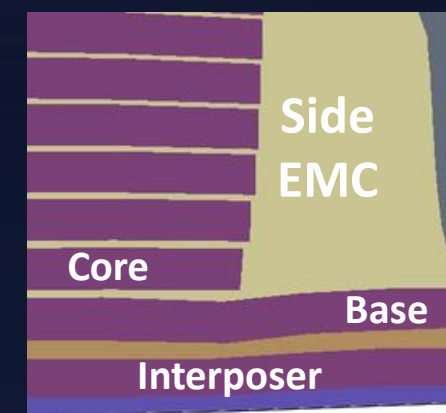
CoWos-L



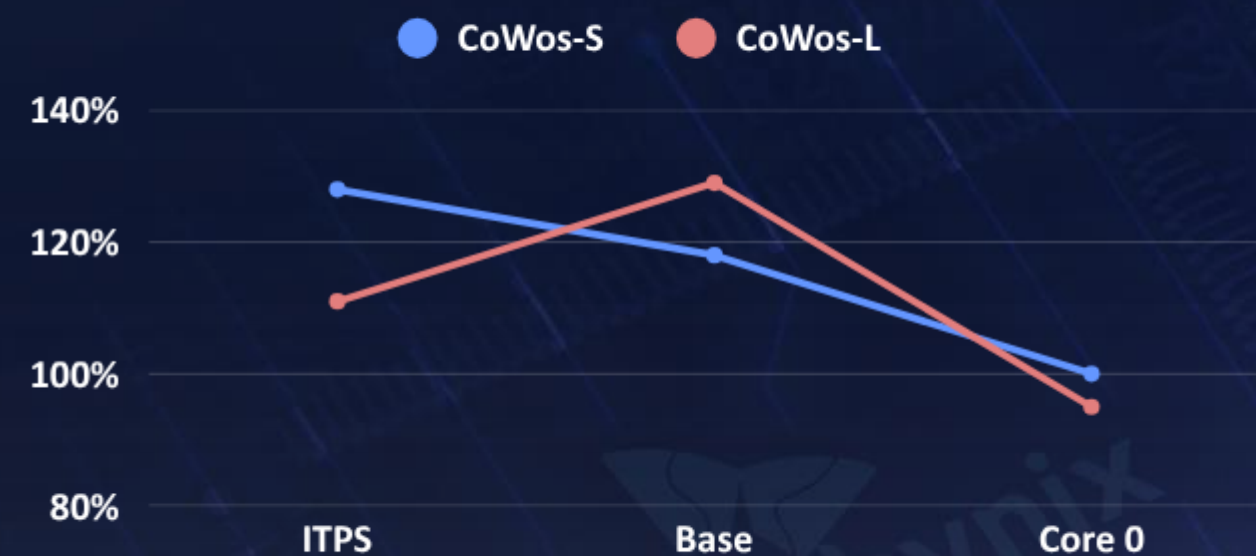
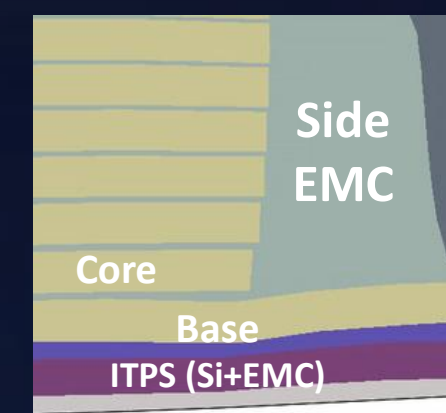
EMIB



CoWos-S



CoWos-L



Stress on HBM/Interposer vs. Adv. PKG technologies

Advanced PKG Future Direction

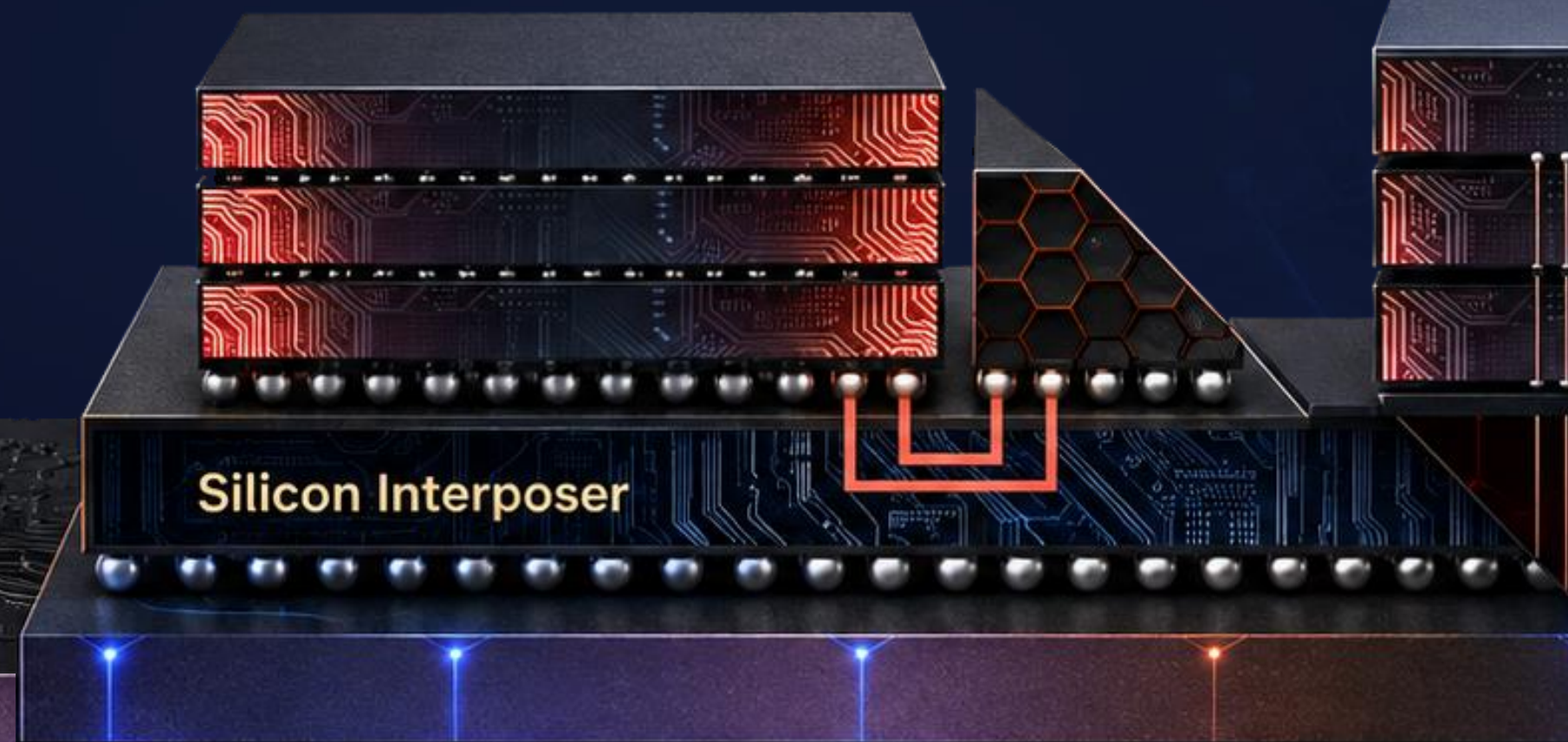
2D Structure

PKG to PKG on Board



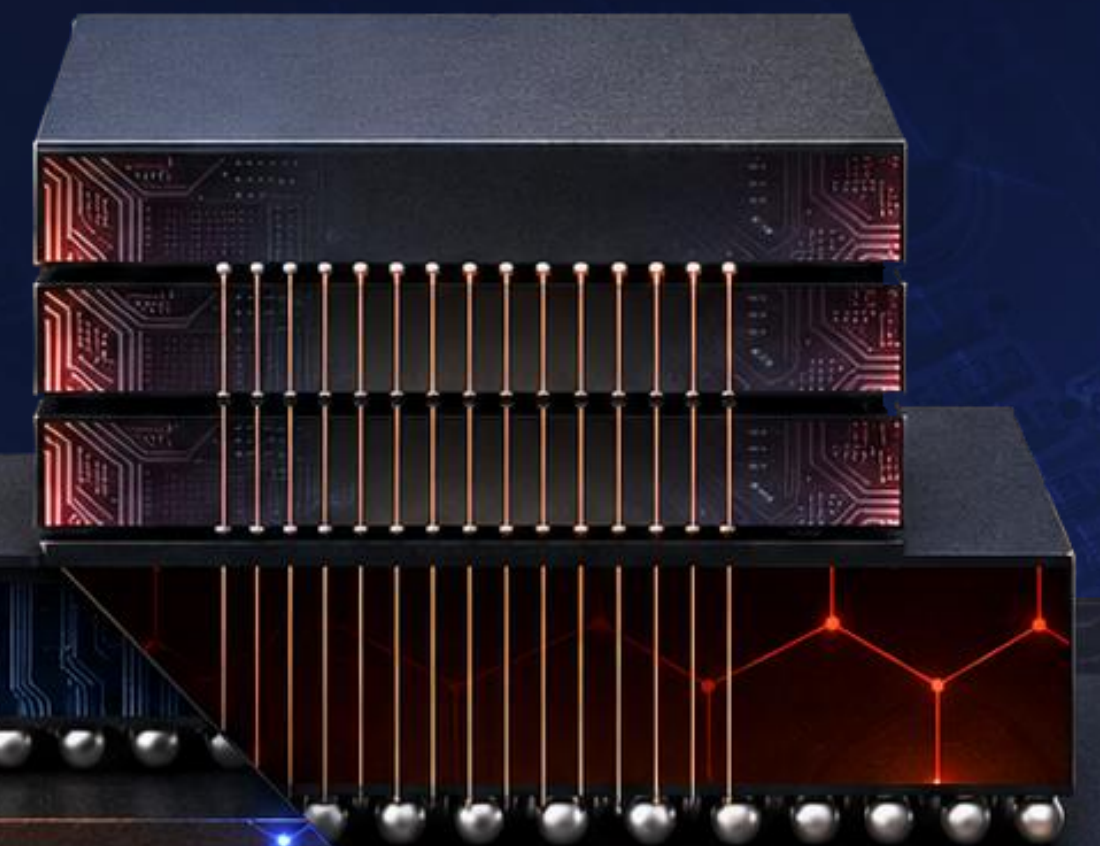
2.5D Structure

Die to Die on Interposer



3D Structure

Die on Die



PCB Substrate



Thank you

